



Transferring decisions to an algorithm: A simple route choice experiment

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ABSTRACT

The future of artificial intelligence is bright and will create new situations in which humans can share or delegate decisions to machines and algorithms – as is likely to be the case in fully autonomous traffic situations. As it is important to understand human-machine interactions and acceptance of these new technologies, we took a deeper look into delegation behavior to an algorithm and its links to personality measures. To this end, we ran the route choice game from Selten et al. (2007), in which 18 players had to choose to either use the main or the side road for 200 periods, but had the opportunity to delegate this decision to a reinforcement algorithm. Results from 216 participants show that delegation rates average around 10% but vary greatly between individual players and with the level of traffic information. Focusing on personality factors, extraverted individuals were robustly found to be more likely to delegate than introvert players. Altogether, results suggest that there is –depending on several factors– hesitation to delegate decisions that affect both the decision maker and other participants.

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1. Introduction

The current evolution of artificial intelligence –for example through advances in deep learning– will undoubtedly create immense potential for innovation, and the development of ever better autonomous systems is moving rapidly. This progress enables machines to autonomously transform information and carry out tasks on behalf of their human agents, and it enables human agents to delegate decisions to machines and complex algorithms that formerly only had supportive functions (Schaefer, Chen, Szalma, & Hancock, 2015). The development of technology that does not need to be supervised or monitored by human agents will create plenty of new situations and possibilities. It especially promises progress on the path to autonomous traffic, as artificial intelligence and machine learning algorithms can be used as tools which empower vehicles to make decisions without human interventions. The idea of self-driving cars has recently generated great interest, and most major car companies are now involved in some kind of research and development of automated vehicles (Gogoll & Uhl, 2018). The inevitable development of autonomously driving vehicles will undoubtedly change the traffic of the future, challenge traditional concepts of mobility and promise substantial gains in human welfare through improved efficiency,

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facilitated mobility, increased safety and environmental benefits. Thus, ever-more capable autonomous systems will certainly have far-reaching societal implications and challenges.

It is therefore important to understand individual behaviors in new situations of sharing or delegating decisions to machines and algorithms. Several studies have focused on different aspects of behavior towards machines as well as on the relationship between human agents and automated systems.

Several of these studies suggest that in situations of human-machine-interactions computers are actually treated similarly to humans. Nass and Moon (2000), for example, reviewed a series of experimental studies and concluded that humans “mindlessly apply social rules and expectations to computers” (p.81). They found that individuals use human social categories and show social behavior towards automated systems. Other studies confirmed that humans tend to behave socially towards computers (Katagiri, Nass, & Takeuchi, 2001; Krämer, Von Der Pütten, & Eimler, 2012; Nass, Fogg, & Moon, 1996; Reeves & Nass, 1996; Riedl, Mohr, Kenning, Davis, & Heekeren, 2011) and view computers as at least partially responsible for their actions (Bechtel, 1985; Eysel & Kuchenbrandt, 2012; Moon, 2003). Strobel and Kirchkamp (2017), for example, drew on a popular economic experiment and tested whether the type of partner – human or machine – affects the participants’ perceptions and choice in a modified dictator game in which the payoff depended on the simultaneous choice of two dictators. The second dictator was either a human or an algorithm which resembled human behavior. They found that perceptions of responsibility differed whereas, just as expected by the theory of mindless application of social rules, actual choices did not depend on whether they played with a human or a computerized agent.

Other studies, however, stress that this observed tendency to apply social norms is weak and depends on many other factors, indicating that differences in behavior towards humans and computers remain (see de Melo, Marsella, & Gratch, 2018). de Melo, Marsella, and Gratch (2016) found that human agents behave very differently when playing the dictator game with a machine than with a human. When playing the ultimatum game, dictators were found to offer substantially more money to a human partner than to an automated receiver and were less likely to perceive guilt when interacting with the machine.

Despite the growing amount of literature on the use of automated aids¹ and on human interactions with automated systems, studies focusing on delegating decisions to autonomous systems are still comparably scarce, even though the agents’ willingness to leave decisions that impact both the agent and third parties to a complex algorithm (whose mode of operation might resemble a black box) will be an important issue in the future, especially in the context of autonomous traffic.² Reservation to accept new technologies could, therefore, hinder their implementation or delay potential benefits. One mentionable study by Gogoll and Uhl (2018) has tested whether people would delegate a decision that affects a third party to a human or to an algorithm that resembles human behavior. They found that the majority of subjects were unwilling to delegate their decision to the algorithm and that observers viewed delegations to the machine quite critically.

Another stream of literature, which is more located in the field of applied psychology, has focused on technology acceptance and its link to different personality factors. Ultimately, factors based on the five-factor model of personality turned out to be useful predictors for technology acceptance, which will be further discussed in a later chapter of this paper. The majority of these studies, however, have focused more on the effect of personality factors on the intention to use a new technology and less on actual behavior towards innovations (e.g. Aldás-Manzano, Ruiz-Mafé, & Sanz-Blas, 2009; Devaraj, Easley, & Michael Crant, 2008; Svendsen, Johnsen, Almås-Sørensen, & Vittersø, 2013). For these reasons, we include personality factors and investigate their link to actual observed delegation behavior in an experimental setting.

Thus, in order to shed some light on this trending but still underexplored issue of delegating decisions to a machine which might be especially prominent in future traffic scenarios, we draw on previous work and replicate a laboratory route choice experiment from Selten, Chmura, Pitz, Kube, and Schreckenberg (2007), which focuses on a generic two-route scenario, has a large number of periods and includes different information treatments. While similar route choice experiments have been investigated in the literature (see for example Chmura & Pitz, 2006; Hartman, 2012; Helbing et al., 2005; Iida, Akiyama, & Uchida, 1992; Mahmassani & Liu, 1999; Schneider & Weimann, 2004), we extend the replicated scenario by including the possibility to delegate the route decision, which affects the player as well as other players, to an reinforcement algorithm. As information is an important factor in traffic behavior (see for example Avineri & Prashker, 2007; Ben-Elia, Di Pace, Bifulco, & Shiftan, 2012; Ben-Elia & Avineri, 2015; Ben-Elia & Shiftan, 2010; Lu, Gao, Ben-Elia, & Pothering, 2014; van Essen, Thomas, van Berkum, & Chorus, 2016), this experiment has two treatments with different information in order to investigate if and how this difference will influence the observed behavior.

2. Methods

2.1. Experimental setup

Just as in the original experiment from Selten et al. (2007), participants were instructed to make a choice between a main road *M* and a side road *S* in order to travel from *A* to *B* (see Fig. 1). The participants’ payoffs depend on their respective travel time needed.

¹ Dzindolet, Pierce, Beck, and Dawe (2002), for example, have created a framework to predict automation use.

² Studies focusing specifically on human-machine interactions in traffic situations are still rare. While some studies focus on the impact of humanlike features on the trust in autonomous vehicles (Waytz, Heafner, & Epley, 2014) or on ethical dilemmas in autonomous traffic situations (e.g. Gogoll and Müller, 2017; Goodall, 2014), not much attention has been paid to an empirical assessment of the road users’ acceptance of autonomously driving systems.

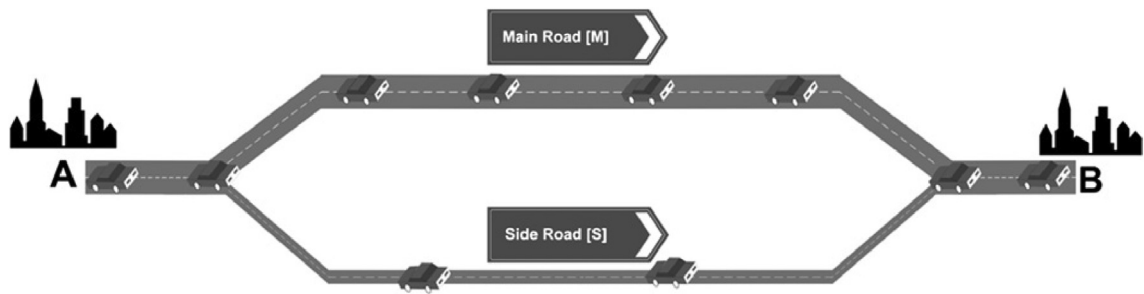


Fig. 1. Experimental setup (Selten et al., 2007, p. 396).

This particular route choice game can be viewed as a coordination problem as players have to coordinate their route choices in order to reach equilibrium. Yet, it is different to classic coordination games in experimental economics as these are characterized by an equilibrium at which every player chooses the same pure strategy. Another kind of similar experimental games are market entry games (see for example Camerer & Lovo, 1999; Duffy & Hopkins, 2005; Rapoport, 1995; Rapoport et al., 1998) in which players have a choice to enter a market or to stay out and where payoff decreases with the number of entering players. Therefore, route choice games can be considered similar to a market entry game with two markets instead of one and without the possibility to stay out of both markets. Market entry games, just as this route choice game, typically have a large number of Nash equilibria and pure equilibria involve considerable coordination on asymmetric outcomes where some agents enter and some stay out, or some take the main road and some the side road, respectively (Duffy & Hopkins, 2005). This class of games has typically found a quick learning towards the equilibrium with subjects switching between strategies. Continued switching is found to impair performance and payoff. However, previous experiments, like the replicated Selten et al. (2007), also showed that additional information reduced switching behavior (for an extensive review please see Ben-Elia & Avineri, 2015).

Coordination games and market entry games have been typically played over a small number of periods. As it has been shown that games with more than 100 rounds produce different behavior and as traffic situations are best described as frequently repeating route choices, this experiment runs over a total of 200 periods (see Selten et al., 2007).

In each session of 200 periods 18 students from the Rhine-Waal University, Germany, were asked to make the above mentioned route choice. They were shown Fig. 1 and explained that if the same number of people travel on the side and on the main road, then M is faster than S (for instructions see Appendix A). More specifically, the travel time, t_M and t_S , depend on the number of participants on M , n_M , and on S , n_S , respectively:

$$t_M = 6 + 2n_M \quad (1)$$

$$t_S = 12 + 3n_S \quad (2)$$

The payoff per period was $40 - t$ with $t = t_M$ if M was chosen and $t = t_S$ if S was chosen. The total payoff per participant was an initial lump sum endowment of 200 experimental currency units (ECU), plus the sum of all 200 period payoffs with a fixed exchange rate of 0.005€ per ECU. Altogether, on average participants earned 2047 ECUs (=10.24€) in the experiment. For answering a consecutive personality questionnaire participants received another 4€. In total, each session took about 1.5 h.

The pure equilibria of this game are at

$$n_M = 12 \text{ and } n_S = 6$$

with a resulting period payoff of 10 ECUs per participant adding up to 180 ECUs per period.³

Just as in the original route choice game, two treatments with different information have been investigated as it is expected that additional information leads to a different travel behavior and possibly even to a different willingness to delegate to the computer. Both treatments consist of six sessions, resulting in a total of 216 different participants. In the control treatment, participants received feedback after every period about their last chosen route, the travel time of the chosen route, their payoff of the last period in ECUs, their cumulative payoff in ECUs and the number of the current period. In the full information treatment, players received additional feedback about the travel time on the non-chosen route in the last period.⁴

Different to the original experiment and regardless of the treatments, however, subjects had the possibility to delegate their decision to the computer C that decided according to a learning algorithm which simulates real human decision making in the given situation. The used algorithm can be best described as a reinforcement learning model –the payoff-sum model– that updates the probabilities with which the main or the side road is chosen. These probabilities are proportional to their

³ This differs from the Pareto-optimum of 181 ECUs. The Pareto-Optimum could be reached with $n_M = 11,4$ and $n_S = 6,6$. This route choice game also has mixed equilibria, including one symmetric mixed equilibrium in which players choose the main road with a probability of $p = 0.6824$. In this symmetric mixed equilibrium 12.2824 players are expected on the main and 5.7176 on the side road.

⁴ The experiment was programmed in oTree (Chen, Schonger, & Wickens, 2016).

propensities which are exogenously determined. In period 1, the propensity to choose the main road and the propensity to use the side road are equal; thus, the probability to choose the main and side road is 0.5 for each player, respectively. In the control treatment, whenever a road is used and the resulting payoff is positive, it is added to that road's propensity. If the payoff is negative, however, its absolute value gets added to the other road's propensity. Thus, if all payoffs were positive, then the propensity is the sum of all previous payoffs for this road plus its initial propensity. In the full information treatment, propensities get updated for both roads in each round. This payoff-sum model was chosen as it is straightforward and has only a minimum number of exogenous parameters. It was also shown in [Selten et al. \(2007\)](#) that the chosen reinforcement model gives a good prediction not only of the mean number of participants on the two roads but also of the fluctuations and number of road changes, and was found to take decisions in a way that resembles human behavior.

2.2. Explanatory factors

In order to understand the individual delegations decisions better, all players were asked to answer an additional questionnaire which included measures of personality. The personality model is based on the factors Openness, Conscientiousness, Extraversion, Agreeableness and Neuroticism (and thus sometimes called OCEAN) which include a huge variety of personality attributes.⁵ Altogether these five factors “capture the essence of one's personality” ([McElroy, Hendrickson, Townsend, & DeMarie, 2007, p. 810](#)). Therefore, we used the 100-item IPIP NEO inventory which is made available on the broad and comprehensive *International Personality Item Pool* (IPIP; [Goldberg et al., 2006; Goldberg, 1999](#)). The inventory is based on [Costa and McCrae \(1992\)](#) NEO Personality Inventory – Revised (NEO PI–R) which is considered by many psychologists to be the best inventory for measuring traits within the five-factor model of personality.

Ultimately, the survey included 20 items per personality factor.⁶ All items were measured with a five-point Likert scale ranging from 1= “Very Accurate” to 5= “Very Inaccurate”, including a neutral option labelled as “Neither Inaccurate nor Accurate”, and reverse scored as necessary. As recommended, the individual item scores were summed up to the respective scale. For representation purposes we normalized the five scales, resulting in 1 being the highest score and 0 the lowest.

Reliability of the measures was checked by assessing the internal consistency through Cronbach's alpha and average interitem correlations ([Briggs & Cheek, 1986](#)) and constructs found to be very reliable (see [Table 1](#)). Cronbach's alpha is consistently high, indicating a very good fit for the use as a research tool ([Nunnally, 1978; Streiner, 2010](#)) and the interitem correlations for this study's scales show acceptable homogeneity⁷: items are related but not too similar and thus show internal consistency without being redundant.

These five factors were included as they are expected to play a significant role in accepting new technologies. *Openness to experience*, for example, is generally linked to curiosity and originality, the motivation to learn and the appreciation of art ([Barnett, Pearson, Pearson, & Kellermanns, 2015](#)). Players scoring high on openness to experience generally tend to be broad-minded, willing to try out new activities and are more likely to accept and embrace change ([Devaraj et al., 2008](#)). Therefore, we expect that the individual openness to experience is positively linked to the willingness to delegate to the computer as this action typically requires these very attributes.

Conscientiousness is characterized by responsibility, thoroughness and the eagerness to achieve. Conscientious people are typically goal-oriented and strive for success, achievement and knowledge and are more willing to take actions to advance ([Digman, 1990](#)). Thus, the relationship between conscientiousness and technology adoption depends strongly on whether participants expect the technology to increase their performance or achievements significantly ([Lane & Manner, 2011](#)). As the algorithm in the experiment simulates human behavior, we do not expect that conscientious players expect it to be better achieving and do not use it more often than other participants.

Extraversion is the degree to which a person is sociable, assertive, initiative, active and enthusiastic. Unlike introverts they enjoy social interaction and find energy and reward in spending time with other people. They generally appreciate activity, and often behave in order to attract social attention ([Ashton, Lee, & Paunonen, 2002](#)). They desire a high social status which is often seen as a reason to adapt new technologies ([Devaraj et al., 2008](#)). As this algorithm used in the route choice game does not generate social interaction – different to many other new technologies – but rather takes away the only means to interact with the players, we do expect a significant negative relationship.

Agreeableness is the tendency to be kind, trusting, cooperative, tolerant and forgiving ([Barrick & Mount, 1991](#)). It is especially expected to influence technology use positively when the technology under consideration fosters cooperation and increases the opinions of others about one, and negatively when it is less helpful in fostering collaboration or reduces the individuals' status ([Devaraj et al., 2008](#)). Thus, in our experiment we do expect a direct relationship to the players' delegation decision, but are uncertain about the direction it takes.

⁵ These five factors have proven to consistently provide a summary of individual personality differences and to be of great reliability and stability across different samples ([Costa & McCrae, 1992; Digman, 1990](#)). Due to its strong predictive validity the five-factor model has been used extensively in the field of psychology ([John et al. 2008](#)), also in explaining technology acceptance ([Devaraj et al., 2008; Lane & Manner, 2011; McElroy & Townsend, & DeMarie, 2007; Svendsen et al., 2013](#)). Accumulated research provides further evidence for the robustness of the five factors ([Barrick & Mount, 1991](#)).

⁶ Ten of these were positive keyed, the other ten negative keyed. Furthermore, items from one scale were not listed together but distributed throughout the questionnaire. This distribution of items and the use of positively and negatively worded items should encourage participants to pay closer attention.

⁷ Average interitem correlations – as another indicator for internal consistency reliability – are generally considered to optimally range between 0.2 and 0.4 ([Milfont & Gouveia, 2006; Piedmont, 2014](#)).

Table 1

Internal consistency reliability measures for the five factors.

	Mean	Standard deviation	Cronbach's Alpha	Average interitem correlation
Openness	0.535	0.180	0.836	0.219
Conscientiousness	0.535	0.197	0.879	0.271
Extraversion	0.521	0.177	0.896	0.356
Agreeableness	0.608	0.181	0.823	0.188
Neuroticism	0.434	0.181	0.877	0.302

Neuroticism is commonly associated with the tendency to experience negative feelings like anxiety, depression, worry, fear and insecurity (Barrick & Mount, 1991). In general, individuals scoring high on neuroticism are less likely to engage in new activities and more likely to perceive a new situation as stressful. The potential effect on technology adaptation, however, is twofold: If neurotic respondents perceive the decision-making in the experiment as stressful, then we expect them to be more likely to delegate to the computer. If they perceive the delegation to the computer, however, as stressful, then an adoption of this technology seems less likely (Devaraj et al., 2008).

In addition to the personality measures, the survey collected information on age and gender as well as on the self-stated willingness to take risks and on individual patience. As trusting an unknown algorithm could be seen as risky, we expect risk-averse participants to be less willing to delegate their decision. Participants with less patience are expected to be more willing to delegate their decision in order to reduce the time spent on considering the best choice themselves.

3. Experimental results

In order to assess the results of this route choice game, we look at different factors discussed in Selten et al. (2007), compare our results with the original paper and extend the findings by analyzing the willingness to delegate decisions and its link to other explanatory factors.

3.1. Route choices

Table 2 shows that the overall mean number of participants on *S* is extremely close to the equilibrium prediction. On average, 6.00 participants used *S* in the control treatment and 6.02 in the full information treatment.

Fig. 2 depicts a typical session of the control treatment and demonstrates, together with Table 2, substantial fluctuations around the theoretical equilibrium, which can be measured by the standard deviation of the number of participants on *S* which is between 1.27 and 2.04. Table 2 also shows that the observed fluctuations are larger under the control treatment than under the full information treatment which turns out to be a significant difference. The null-hypothesis of no difference is rejected by a Wilcoxon-Mann-Whitney test on the 1% level.

These findings are perfectly in line with the original ones by Selten et al. (2007) who argued that the observed fluctuations can be easily explained by the multiplicity of pure strategy equilibria which poses a coordination problem.⁸ However, in this paper, a look at the number of participants using the side road does not allow us to draw conclusions about the actual choices of participants. The use of the side road could have been a decision by the player or by the algorithm. Therefore, Table 2 also depicts the number of players that delegated their decision to the computer.

When taking a closer look at actual decisions and the possibility to delegate the decision to the learning algorithm, we find that on average 1.86 participants per period chose the computer, more precisely 2.73 in the control treatment and 1.30 in the full information treatment. The Wilcoxon-Mann-Whitney test confirms that, interestingly, participants chose the computer *C* significantly more often in the control treatment than in the full information treatment (see also Table 1). Additionally, we find that the number of players willing to delegate the decision to the algorithm decreases slightly over periods (see Fig. 3).

The Spearman-rank-correlation between the number of players choosing *C* and the number of periods, however, is stronger in the control treatment ($\rho = -0.2959$) than in the full information treatment ($\rho = -0.0982$) which might result from the fact that in the latter treatment less players chose to delegate to the computer from the beginning. When looking at it from another angle, we find that on average players delegate to the computer in 22 out of 200 rounds. Fig. 4 further emphasizes that the distribution of the total number of delegations is heavily right-skewed: a quarter of the participants never delegated to the computer, whereas only 4.6% chose the computer in at least half of their rounds. Rather unexpectedly, two players trusted the algorithm in every single round.

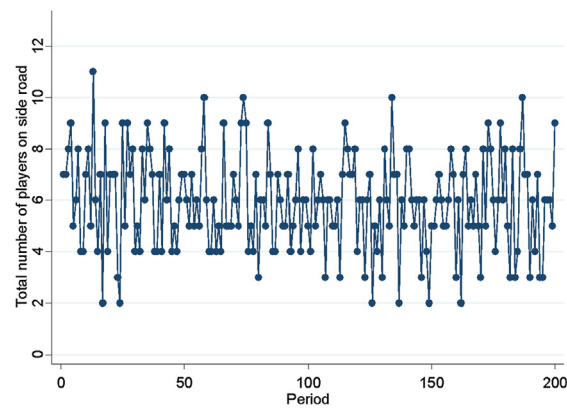
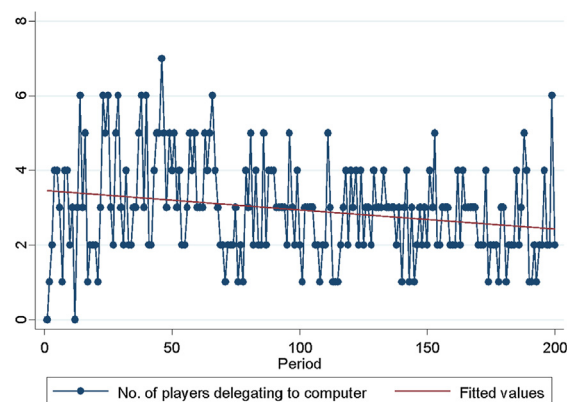
Of course, there is also a highly significant difference between the treatments when focusing on the average number of delegations to the algorithm: Whereas an average player in the control treatment delegates the decision to the computer in 30.3 out of 200 periods, an average player in the full information treatment only does so in 14.4 periods.

⁸ Even though the equilibrium number of participants on the side road is 6, it could be any set of 6 players which causes the appearance of several equilibria.

Table 2

Number of players on side road S and number of players delegating to computer C.

		Number of players on S		Number of players delegating	
		Mean	std. dev.	Mean	std. dev.
Control treatment	session I 01	5.95	1.558	2.72	1.294
	session I 02	6.01	1.779	2.53	1.081
	session I 03	5.91	1.875	2.59	1.460
	session I 04	6.06	1.774	2.20	1.460
	session I 05	6.03	1.565	3.40	1.385
	session I 06	6.08	2.037	2.94	1.337
	Total	6.00	1.774	2.73	1.345
Full information treatment	session II 01	6.00	1.380	1.51	0.860
	session II 02	6.03	1.269	0.71	0.747
	session II 03	5.95	1.667	1.84	1.032
	session II 04	5.96	1.638	1.21	0.892
	session II 05	6.28	1.828	1.74	0.802
	session II 06	5.92	1.389	0.78	0.784
	Total	6.02	1.545	1.30	0.964

**Fig. 2.** Number of players on side road S in a typical session of the control treatment.**Fig. 3.** Number of players delegating their decision to the computer in a typical session of the control treatment.

3.2. Changes in route choices

The observed high fluctuations suggest that participants change their decisions, and thus switch roads, quite often. Table 3 shows the mean and standard deviation of road and choice changes, where 'road' is the final road taken (S or M), 'choice' is the actual decision made by the player (S, M, or C) and 'delegation switches' indicates whether the participants switch

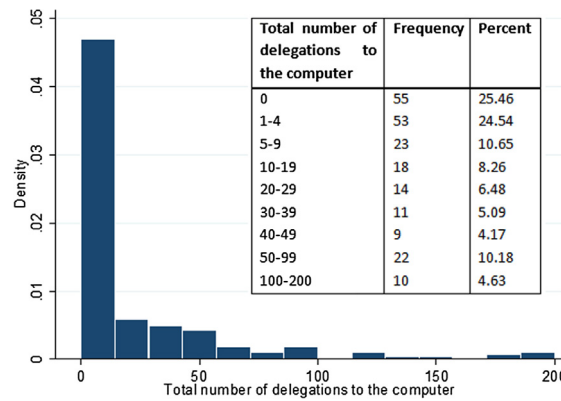


Fig. 4. Distribution of players' total number of delegations to the computer.

Table 3
Session overview.

		Number of road changes		Number of choice changes		Number of delegation switches	
		Mean	std. dev.	Mean	std. dev.	Mean	std. dev.
Control treatment	session I 01	4.22	1.996	4.70	1.983	1.55	1.627
	session I 02	5.09	1.750	5.95	1.945	2.10	1.717
	session I 03	6.57	1.912	7.98	2.237	2.98	2.118
	session I 04	5.81	2.364	6.33	2.483	1.73	1.749
	session I 05	4.44	1.930	5.23	2.332	2.52	1.990
	session I 06	6.20	2.343	7.19	2.421	2.46	1.822
	total	5.39	2.240	6.23	2.503	2.22	1.908
Full information treatment	session II 01	3.27	1.551	3.64	1.707	1.66	1.623
	session II 02	2.72	1.376	2.92	1.513	0.95	1.539
	session II 03	5.25	1.846	5.63	1.759	1.97	1.565
	session II 04	4.12	1.969	4.43	2.118	1.52	1.540
	session II 05	6.67	2.339	6.25	2.299	0.75	1.459
	session II 06	4.67	1.937	4.64	1.968	0.63	1.464
	total	4.45	2.271	4.58	2.217	1.25	1.611

between delegating and not delegating (C or no C). It reveals significant differences between the treatments. Whereas the mean choice change in the control treatment is 6.23 (road change: 5.39, delegation switches: 2.22), it is 4.58 (road change: 4.45, delegation switches: 1.25) in the full information treatment. Again, the null-hypothesis of no difference is rejected by a Wilcoxon–Mann–Whitney test on the significance level of 1%. Our findings again confirm previous results from the original paper.

But not only do the mean numbers of road and choice changes as well as delegation switches, which are inevitably linked to each other anyway, differ significantly between the treatments, they also slightly decrease over time. Fig. 5 further shows that the number of road changes per period for a typical session of the control treatment decreases over time. Indeed, we find a negative trend for all sessions in the control treatment and nearly all sessions in the full information treatment (here two sessions show an indifferent trend for both the number of road and choice changes).

Focusing on delegation behavior, we find that there is a positive link between the player's number of delegations and his number of road changes in the full information treatment. This finding, however, could be attributed to the updating mechanism of the algorithm.

3.3. Payoffs

Average period payoffs were significantly lower in the control than in the full information treatment. Taking a closer look, we find that the number of road and choice changes is also negatively correlated with the subject's cumulative payoff in both treatments – a finding which again confirms original results. Additionally, this correlation is slightly stronger in the full information than in the control treatment.⁹

⁹ The Spearman rank correlations between cumulative payoff and the number of road and choice changes are strictly negative (route change: control treatment: Spearman's rho = −0.6705, full information treatment: Spearman's rho = −0.7513; choice change: control treatment: Spearman's rho = −0.5730, full information treatment: Spearman's rho = −0.7318).

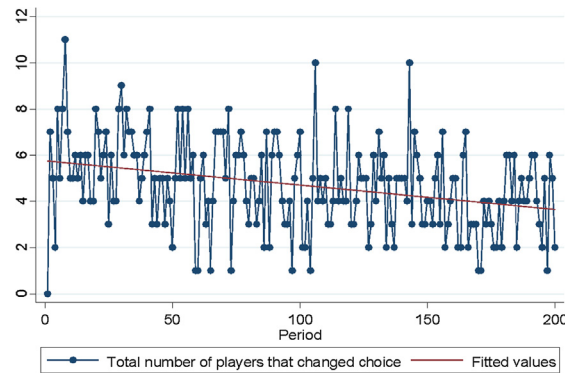


Fig. 5. Number of participants that changed their choice per period in a typical session of the control treatment.

Therefore, we can argue that the average period payoff (and thus the cumulative payoff) is higher in the full information treatment as people switch the road or change their decision less often. As we focus especially on players' delegations to the computer we also look at their relationship to the cumulative payoff (see Fig. 6). Interestingly, we find a slightly negative relationship. The more the players delegate, the lower their final payoff is. When looking at the two treatments separately, however, this relationship can only be found for the full information treatment and is most likely linked to the finding in 3.2 that delegations resulted in more road changes and more road changes in a lower payoff.

3.4. Determinants of delegation behavior

The previous findings suggested that the delegation decision is linked to different factors of the experiment. But when exactly do participants decide to delegate or to switch between delegating to the computer and making own decisions instead? In order to answer these questions, we take a closer look at two binary dependent variables: whether the participant delegated his decision in the round or not (=1 yes, =0 no), and whether the player ends a sequence of delegating or not delegating to the computer (=1 ends a sequence of delegating or not-delegating and switches to the opposite strategy, =0 does not switch delegation strategy). Therefore we use basic logistic panel regressions with either of the dependent variables and several independent variables (see Table 4).

Regression results indicate several significant factors: It is found, for example, that the lower the player's period payoff, the more likely the player is to end the sequence of delegating or not delegating, thus switching in the next round. Unsurprisingly, players earning zero ECUs are 11% more likely to switch in the next period than players who earned 20 ECUs. Furthermore, the propensity to switch between delegating and not delegating decreases with the total number of players on the side road and increases with the total number of players that delegate the decision to the computer. Previous behavior is also a good indicator: the higher the number of the sequence (thus the more often the player has switched before) and the shorter the length of the sequence (thus the shorter the time since last switching), the more likely the player is to switch again. A look at predicted probabilities suggests that the probability of switching is around 37% higher for a player who has already switched 150 times than for one who has previously switched 50 times.

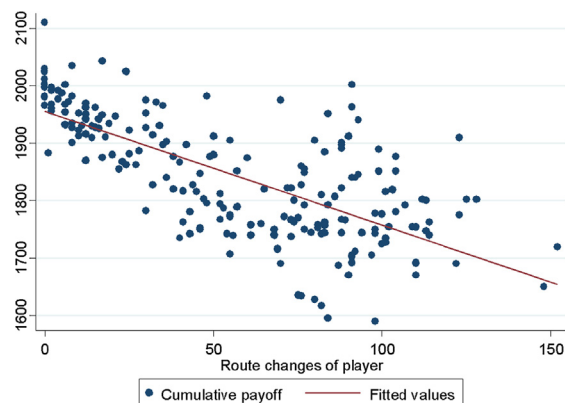


Fig. 6. Individual road changes and cumulative payoff.

Table 4
Results from logit regression.

	Delegation (i)	Switch delegation strategy (ii)
Time on non-chosen road, previous round	0.019 (0.021)	0.005 (0.00)
Player payoff, previous round	−0.020 (0.013)	−0.080*** (0.00)
Number of players on S, previous round	−0.056* (0.033)	−0.060*** (0.01)
Number of sequence	0.018*** (0.003)	0.022*** (0.00)
Length of sequence	−0.016** (0.007)	−0.035*** (0.00)
Full information treatment	−0.586** (0.275)	0.170*** (0.04)
Constant	−0.974** (0.408)	−0.578*** (0.17)
Pseudo ¹ R ²	0.117	0.212
Number of observations	42,984	42,984

Notes: The dependent variable in (i) is a binary variable (=1 if player delegated to computer, =0 otherwise). The dependent variable in (ii) indicates whether player switched (either from delegation to no delegation or vice versa, =1 if switched, =0 otherwise). Cluster-robust standard errors in parenthesis.

* $p < 0.1$.

** $p < 0.05$.

*** $p < 0.01$.

¹ McFadden's pseudo R² tend to be considerably lower than those of the R² index in a standard OLS regression and it is argued that values of 0.2–0.4 “represent an excellent fit” (McFadden, 1978, p. 306). Other sources see these values rather as a good fit (e.g. Domencich and McFadden, 1975).

Furthermore, we investigate whether the effect of the investigated variables depends on other variables. A look at these interaction effects shows few significant and relevant interactions. It reveals interesting insights about the effect of generated payoffs on individual delegation behavior (see Fig. 7). The left graph, for example, shows that the probability to delegate to the computer depends strongly on the payoff received in the previous round in the full information but less so in the control treatment.

The right graph shows that the payoff's average marginal effect on switching the delegation strategy depends on the length of the sequence. This makes sense: the longer a player sticks to her delegation strategy the less sensitive she is to different payoffs.

4. External factors and their influence on delegation decisions

While the previous results have focused on behavior in the experiment only, insights from the consecutively conducted questionnaire will be added in order to further investigate the link between personality measures and observed delegation behavior.

4.1. Regression results

In order to get a more comprehensive impression of the link between personality measures and delegation decisions we return to logistic panel regressions using the two binary dependent variables from Section 3.4 (see Table 5).¹⁰

Model (i) and (ii) show that the personality factor extraversion is significantly and negatively linked to the player's delegation decision and robust to the inclusion of several control variables. The size of the effect is not negligible: A further look at predicted probabilities indicates that a participant in the lowest decile of the extraversion distribution (and thus being someone among the most introverted) is 15% more likely to delegate to the computer than someone in the highest decile (someone among the most extraverted people). This effect weakens and turns less significant once the switch between delegation decisions becomes the dependent variable in model (iii) and (iv). Here, however, it turns out that there is a significant positive link between the personality factor conscientiousness and our dependent variable. The effect is, however, quite small as a player in the highest decile of the conscientiousness distribution is only 2% more likely to end a sequence of same delegation decisions than a player in the lowest decile.

Again, we take into consideration that variables might not only directly affect individual delegation decisions but that the effect might depend on other variables. We checked for interactions and found that the gender of a participant acted as a moderating factor for two personality factors: Fig. 8 highlights that there is no effect of conscientiousness on the probability

¹⁰ Thus, the players' decision to delegate or not delegate in each round as well as the players' decision to switch between delegating and not delegating.

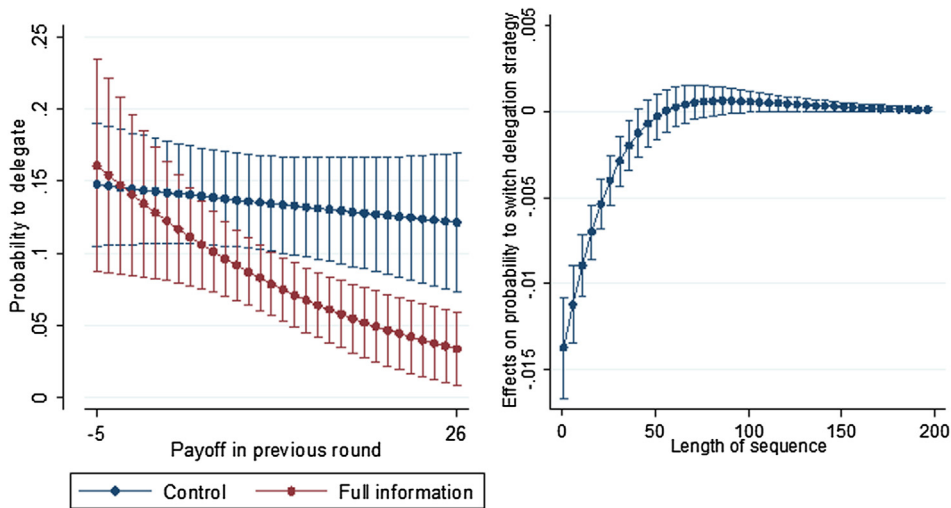


Fig. 7. Interaction effects. Left side: Probability to delegate by payoff in previous round and treatment, Right side: Average marginal effects of payoff by length of sequence.

Table 5
Personality measures and delegation decisions.

	Delegation		Switch delegation strategy	
	(i)	(ii)	(iii)	(iv)
Openness	−0.141 (0.86)	−0.591 (1.00)	0.342 (0.50)	0.145 (0.21)
Conscientiousness	1.392 (0.96)	1.20 (1.07)	0.920 (0.56)	0.539** (0.26)
Extraversion	−2.026** (1.02)	−1.731** (0.92)	−1.140 (0.71)	−0.455 (0.27)
Agreeableness	−1.179 (0.93)	−0.856 (0.68)	−0.415 (0.75)	−0.110 (0.30)
Neuroticism	0.207 (0.97)	0.688 (1.16)	−0.326 (0.67)	−0.302 (0.25)
Age		0.028 (0.02)		−0.015** (0.01)
Gender		0.150 (0.26)		−0.008 (0.08)
Risk		−0.000 (0.11)		−0.024 (0.02)
Patience		−0.045* (0.02)		0.007 (0.02)
Player payoff, previous round		−0.009 (0.01)		−0.078*** (0.00)
Number of players on S, previous round		−0.026** (0.01)		−0.056*** (0.01)
Full information treatment		−0.462* (0.25)		−0.157* (0.09)
Number of sequence		0.018*** (0.00)		0.021*** (0.00)
Length of sequence		−0.016** (0.01)		−0.035*** (0.00)
Constant	−3.335** (1.61)	1.081 (2.09)	−1.947** (0.81)	0.002 (0.43)
Pseudo R ²	0.025	0.139	0.012	0.2112
Number of observations	43,200	42,586	43,200	42,586

Notes: The dependent variable in (i) and (ii) is a binary variable (=1 if player delegated to computer, =0 otherwise). The dependent variable in (iii) and (iv) indicates whether player switched (either from delegation to no delegation or vice versa, =1 if switched, =0 otherwise). Cluster-robust standard errors in parenthesis.

* p < 0.1.

** p < 0.05.

*** p < 0.01.

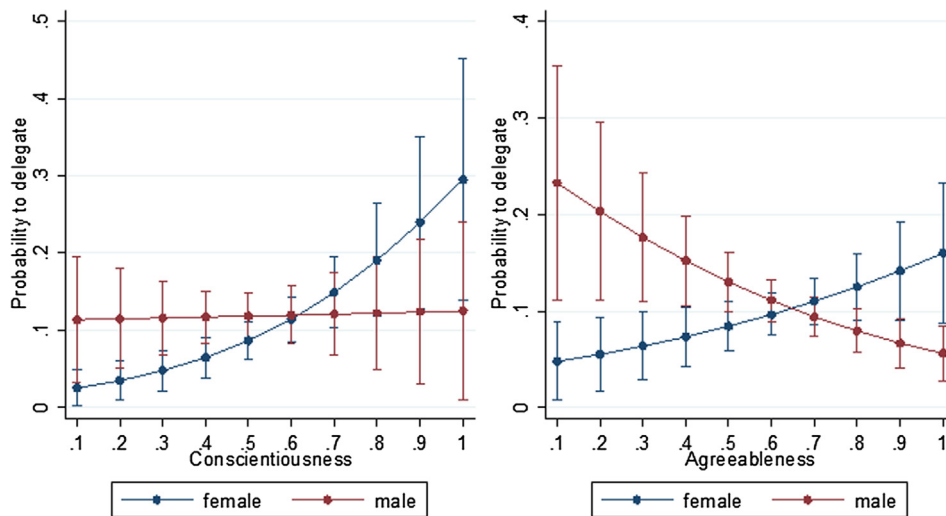


Fig. 8. Interaction effects. Left: Probability to delegate by gender and conscientiousness score, right: Probability to delegate by gender and agreeableness score.

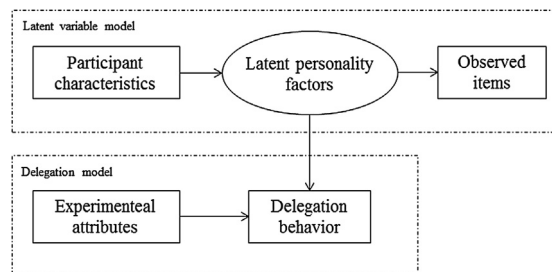


Fig. 9. Structural equation model . adapted from Bekhor & Albert, 2014

to delegate for male participants, but that the female participants' probability to delegate increases with their conscientiousness score. It also depicts an opposing effect of agreeableness for male and female participants. Whereas the probability to delegate increases with the agreeableness score for female participants, it decreases for males.

4.2. Robustness test

As the five personality factors are latent constructs and not directly measured, we use structural equation modeling (SEM) as robustness check of our findings. SEM differs from the previously used traditional regression modeling approach by adopting a multivariate technique which incorporates both observed and latent variables (Ben-Akiva et al., 2007; Hoyle, 1995; MacCallum & Austin, 2000). The inclusion of latent variables in route choice experiments using SEM has been modeled in the literature before, see for a general framework Prato, Bekhor, and Pronello (2012), who have introduced a hybrid discrete choice model that integrates a latent variable model and a route choice model, and Bekhor and Albert (2014) for a two-route choice experiment.

We follow this approach using maximum likelihood procedures, but instead of looking at discrete route choices and their utilities, we continue to focus on delegation behavior (see Fig. 9).

Therefore, the latent variable model consists of structural equations (which link latent variables with individual characteristics) and measurement equations (which link unobservable personality factors with observable items). The delegation model consists of structural equations which link observable and latent variables to the observed delegation behavior. The simultaneous estimation of the model structure produces consistent and efficient estimates of the parameters and results in latent variables that provide the best fit to both the delegation and the latent variables indicators (Ben-Akiva et al., 2007). Given the complexity of a dynamic model we look at the aggregated outcomes and report the effects on the aggregated measure which has been used before (see for example Fig. 4), the total amount of delegations to the computer. Having a look at the latent variable model, we find all 100 items to be significantly¹¹ linked to the respective latent personality measure as well

¹¹ All significant at least at the 0.01% level.

Table 6
Results from SEM.

	Total delegation to computer
Openness	–0.045 (0.82)
Conscientiousness	0.201 (0.09) ^{**}
Extraversion	–0.224 (0.09) ^{**}
Agreeableness	–0.059 (0.09)
Neuroticism	0.002 (0.11)
Cumulative payoff	–0.199 (0.07) ^{**}
Full information treatment	–0.139 (0.07) ^{**}
Root Mean Square Error of Approximation (RMSEA)	0.012
Standardized Root Mean Square Residual (SRMR)	0.072
Number of individuals	216

Notes: Measurement model parameters and latent variable model parameters not shown due to space limitations (as this dataset contains 100 indicators plus several traveler characteristics).

*p < 0.1.

***p < 0.01.

**p < 0.05.

as significant correlations between certain individual characteristics and latent variables. A look at the delegation choice model confirms previous findings (see Table 6).

The model fit is acceptable. The coefficients of the latent personality factors conscientiousness and extraversion are significant and show the expected sign. Both factors stay significant if we include participant characteristics in the delegation instead of the latent variable model. The results confirm the robustness of the previous regressions.

Altogether, external factors played a minor role in delegation decisions, whereas internal factors like the amount of information, previous experience and the length and number of sequence turned out to be robust and significant predictors of delegation behavior. Results nevertheless support that considering latent personality factors alongside traditional variables can enrich the comprehension of delegation behavior.

5. Discussion and conclusion

This study replicated the popular route choice game from Selten et al. (2007), and aimed to extend the game to contribute to a newer branch of literature that focuses on human-machine interactions, the willingness to delegate decision making to a computer and its link to the five major personality factors. Therefore, we ran a route choice game, in which in each of the twelve sessions 18 players had to choose to either use the main or the side road for 200 periods. Their payoff depended on their respective travel time which again was determined by the number of people on the chosen road. It can thus be regarded as a coordination problem in which all players have to coordinate their route choices to reach the equilibrium. Just as in the original paper, two treatments with different information were looked at. The experimental evidence replicates the original findings as well and confirms their reliability: despite fluctuations, participants on average played very close to the equilibrium. These fluctuations, and thus also the number of road changes, were lower in the full information treatment II than in the control treatment, most likely due to the additional feedback given to players in the full information treatment. The findings suggest that participants in the control treatment switch more often, as they need to take the other route from time to time to receive information about both roads.¹² This information seeking is not necessary in the full information treatment as players already receive information on both roads after every period.

When focusing on the main objective, namely on analyzing the players' behavior towards delegating decisions to a learning algorithm, we find that delegation rates were rather low. On average 2.73 players chose to delegate their decision in the control treatment, whereas less than half players chose to do so in the full information treatment. The lack of information and the consequent insecurity about route payoffs might have motivated participants to entrust their decision to the computer. In the full information treatment, on the other hand, participants might have been more confident that they had all the necessary information available to make good decisions themselves. This could indicate that players are especially willing to delegate their decision to an algorithm in cases of lacking information. The more information players have, the more confident they could be to choose the right strategy without the help of a "black box". Overall, people were more likely to change

¹² It might have also been the attempt to reduce travel time and generate higher payoffs. However, the negative correlation between payoffs and the number of road changes suggests that these attempts were rarely successful.

their delegation strategy whenever payoffs were low –and thus probably dissatisfying– when players had switched their strategy frequently before and when players did not stick to that strategy for very long.

The inclusion of personality factors provided evidence that extraversion contributes to individual differences in delegating in this experiment whereas other factors were not found to be robustly significant. Extraverted people were found to be less likely to delegate their decisions to the algorithm than introverted people, possibly because extraverted players appreciate activity and enjoy social interaction more than introverts and might have thus been more hesitant to hand over their only means to directly participate to a computer which is in line with our initial hypothesis. A look at personality factors also showed that conscientious players (those who are more eager to achieve) were more likely to switch strategies than less conscientious participants, possibly because they were trying to improve their performance instead of conveniently sticking to their previous strategy.

All in all, results suggest that individuals are hesitant to delegate to an algorithm whose operating principle is not obvious. Even though the majority of people use automatic aids like navigation systems, handing over actual decisions that impact the whole group involved seems to be less accepted. This acceptance, however, might increase with certain personality types and with the complexity of the situation. The experimental results suggest that individuals are more likely to delegate to an algorithm, and thus to trust a machine, when they do not have all information about the situation and see thus a potential advantage of the artificial intelligence. This willingness to delegate depended significantly on certain personality factors.

Overall, this research contributes significantly to the literature on human-machine interactions by not focusing on behavior towards computerized agents in certain situations but by analyzing the delegation behavior of individuals. Thus, players were not asked to interact with computer controlled players, but to entrust the algorithm with making decisions on their behalf which was rarely considered in previous research. Furthermore, the present research analyzes decisions which have monetary consequences and are thus more relevant to the players than non-incentivized tasks which are especially common in some strands of the literature on human-machine interactions.

Despite these methodological advantages, the research has some limitations which could lead to possible future extensions. Firstly, it has to be kept in mind that there is the possibility that a small change in the algorithm will have a dramatic effect on the delegation rate. A basic reinforcement algorithm, which resembles real human behavior, was used in this study. We could, however, argue that future autonomous systems are likely to perform tasks even better than humans and might thus be perceived differently than our basic algorithm. If an algorithm is associated with a high probability of success –even if the algorithm is not fully optimal– the emergence of higher delegation rates is likely as the research summarized in [Erev and Roth \(2014\)](#) suggests. Future research should therefore test whether the specific algorithm or its presentation and explanation have an influence on behavior towards computerized agents or delegation behavior and how the delegation behavior differs with different algorithms.

Similarly, we would expect that the use of autonomous systems, especially in the context of traffic scenarios, would enable individuals to use their time differently. If people delegated the operation of a car to an autonomously driving system, they could then use the time formerly used to drive the car for different and probably more enjoyable activities. In the experiment, however, players who delegated to the algorithm could generally not make use of the saved time in an enjoyable or meaningful way. Thinking about the best road from A to B or the best strategy to improve earnings might have been more interesting than relying completely on the computer instead. Future research could account for these limitations and improve the present findings that, nevertheless, add to the existing literature.

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Appendix A

A1. Definition and summary statistics of used variables

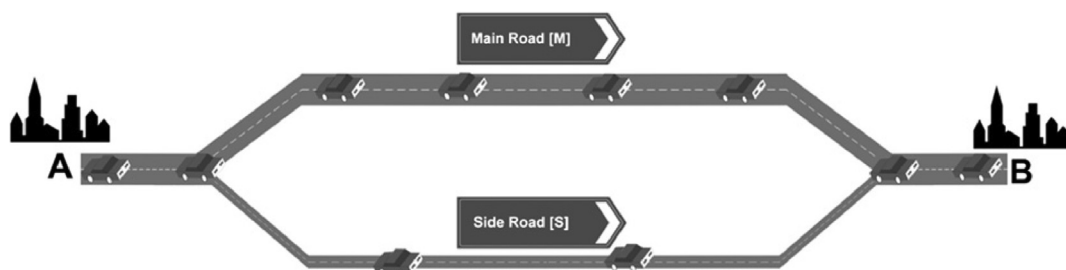
Variable	Definition	Min	Max	Mean	Standard deviation
Openness	20 items per personality factors. Items were measured with a	0	1	0.535	0.180
Conscientiousness	5-point Likert scale ranging from 1 to 5. Individual item scores	0	1	0.535	0.197
Extraversion	summed up to respective personality factor and normalized for	0	1	0.521	0.177
Agreeableness	representation purposes (see Section 2.2).	0	1	0.608	0.181
Neuroticism		0	1	0.434	0.181

Discussion and conclusion (continued)

Variable	Definition	Min	Max	Mean	Standard deviation
Time on non-chosen road	Travel time on non-chosen road	10	54	27.82	5.82
Cumulative payoff	Cumulated payoff generated in experiment, without lump-sum payments	1590	2110	1847.1	105.7
Player payoff in round	Payoff generated in round	−14	26	9.24	3.94
Number of players on S	Total number of players on side road	2	14	6.01	1.66
Number of players choosing C	Total number of players delegating to computer	0	8	2.01	1.37
Total delegations	A player's total number of delegation to the computer	0	200	22.36	38.58
Number of sequence	Number of sequence	1	126	11.64	16.58
Length of sequence	Length of sequence	1	200	53.82	56.19
Full information treatment	=1 If treatment is the full information treatment, =0 if control treatment	0	1	0.5	0.5
Age	Age of participant in years	18	43	24.73	4.53
Gender	=1 if male, =0 otherwise	0	1	0.56	0.49
Risk	Are you generally a person who is willing to take risks or do you try to avoid taking risks? Please choose a value on the scale by using the slider. The value 0 means not at all willing to take risks and the value 10 means very willing to take risks.	0	10	5.99	2.25
Patience	Are you generally an impatient person or someone who always shows great patience? Please choose a value on the scale by using the slider. The value 0 means very impatient and the value 10 means very patient.	0	10	5.98	2.55
Number of road changes	Total number of times a player changed roads	0	191	60.05	43.13
Number of choice changes	Total number of times a player changed his choice	0	152	54.63	38.39

Instructions

- Altogether 18 persons are participating in this experiment. The game situation is the same for every participant. The experiment consists of **200 periods**.
- In each period you are travelling from a **starting point A** to an **arrival point B**. You can either choose a **main road** or a **side road** to get from A to B (see drawing).



- For the travel time from A to B, the following holds: On both routes, the travel time increases with increasing traffic and decreases with decreasing traffic. If traffic is the same on the main and on the side road, the travel time is shorter on the main road than on the side road.
- In every period, you have the choice between **making a route choice yourself or letting the computer make a route choice for you**. The computer decides according to a learning algorithm which simulates real human decision making in the given situation.
- **Your payoffs per period:** After each period you will receive a period payoff P which depends on the travel time T . Hereby holds: $P = 40 - T$, i.e. the shorter the travel time needed, the higher the payoff.
- **Your information per period:**
 - o The travel time on the route that you chose in the preceding period
 - o The travel time on the route that you did not choose in the preceding period (*only in the full information treatment*)
 - o Your route chosen in the preceding period
 - o Your period payoffs in the preceding period in **Talers**
 - o Your cumulative payoffs before the route choice in **Talers**
 - o Number of the current period.
- Each participant receives a seed capital of **200 Talers**. The exchange rate is **0.5 Cent** per Taler.

Independent of your success in the game, you will receive a **4 €** lump-sum payment for answering a survey after the game.

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